Capstone Project: Honey-LLM

1. Introduction

In 2026, the cybersecurity landscape has shifted from static code exploits to 'conversational warfare.' With the mass adoption of Agentic AI, autonomous bots now handle over 80% of digital transactions, outnumbering human users. This has opened a massive attack surface: Prompt Injection and Semantic Chaining. Traditional security measures (firewalls and regex filters) are blind to these attacks because they lack semantic context.

Honey-LLM (The Smart Mirror Trap) is a proactive defense system designed to fight AI with AI. It identifies malicious intent in real-time and, instead of simply blocking the attacker, lures them into a generative sandbox. There, it feeds them 'hallucinated' fake secrets, wasting their resources while simultaneously learning from their techniques.

2. Problem Statement

- Static Defense Failure: Current firewalls cannot distinguish between a legitimate AI prompt and a sophisticated jailbreak attempt.
- The Confused Deputy Problem: Attackers trick trusted AI agents into using their high-level privileges to leak real company data.
- Reactive Lag: By the time a human security analyst detects a breach, autonomous AI agents have already exfiltrated thousands of files at machine speed.

3. Unique Selling Propositions (USPs)

1. Intent-Aware Detection: Uses a specialized ML classifier to detect the 'vibe' of a hack rather than just searching for keywords.
2. Counter-Hallucination Engine: Dynamically generates realistic but fake data (passwords, keys, docs) to maximize attacker dwell time.
3. Automated Guardrail Synthesis: Converts captured attack patterns into protection rules (JSON/NeMo) for the production system instantly.
4. Zero-Trust Sandboxing: Fully isolated environment ensures 'break-out' risks are mitigated through Docker virtualization.

4. Pros and Cons

PROS:
- High Innovation: Addresses the 'Year of the Defender' (2026) shift towards proactive AI security.
- Cost-Effective: Reduces server load by trapping bots in lightweight 'Deception Containers.'
- Rich Intelligence: Captures Zero-Day prompt injection techniques that traditional tools miss.

CONS:

- Technical Complexity: Requires orchestration between detection, generation, and isolation layers.
- Latency Risks: Believable deception requires high-speed inference (SLMs like Llama-3-8B are required).
- Operational Overhead: Requires careful resource management to prevent the honeypot from consuming too much GPU/Memory.

5. Six-Month Roadmap

Month 1: Infrastructure & Sandboxing
- Build Docker-based isolation environment.
- Deploy local SLM (Phi-3/Llama-3-8B) and set up the 'Sarah' Persona.

Month 2: MVP Development
- Implement a basic chat interface and logging system.
- Create 'Bait' files (fake .csv and .env) for the AI to refer to.

Month 3: The Intelligence Layer
- Train/Fine-tune a DistilBERT classifier for malicious intent detection.
- Integrate 'JailbreakBench' datasets for high-accuracy training.

Month 4: Deception Logic
- Develop the 'Liar' system that switches the AI to hallucination mode upon threat detection.
- Program the AI to generate fake passwords and SSH keys on demand.

Month 5: Automated Hardening
- Build the 'Self-Healing' script that turns attack logs into security guardrails.
- Run red-teaming simulations using Garak and PyRIT tools.

Month 6: Finalization & Demo
- Finalize Thesis and Performance Metrics (Detection Rate, Dwell Time).
- Record a live demonstration of an automated bot being trapped in the 'Mirror Maze.'